

APSC 103

Phase 4 – Final Design Report

Team: 529E

Project Manager: Michael Radan

Faculty Advisor: Professor Jim McLellan

Due Date: March 25th, 2024

"We do hereby verify that this written report is our own individual work and contains our own original ideas, concepts, and designs. No portion of this report has been copied in whole or in part from another source, with the possible exception of properly referenced material."

Signed by:

Rafael Costa - XX

Nikhil Naran - XX

Fedor Kim - XX

Daniil Nistribenko - XX

Ray Chen - XX

Executive Summary

Queen's University stands to benefit from the proposed digital student card system, designed to revolutionize campus life. This innovative solution promises students' convenience, enhanced security features, and real-time access to essential campus information such as dining hall occupancy. Importantly, it aligns with the university's sustainability goals by significantly reducing plastic waste associated with traditional student cards.

A functional prototype demonstrates the core features of the digital student card system. First, students must be able to seamlessly use their smartphones for meal purchases, Trade A Meals (TAMs), Flex dollars, and Athletics and Recreation Center (ARC) access. Second, encryption protocols and other security measures must safeguard sensitive student data. Finally, to ensure accessibility for all students, the digital card system must be designed to complement the existing physical card system.

The project prioritizes enhancing the student experience while streamlining campus operations for staff. Development of the prototype utilized Java Swing for the front-end user interface and a Comma Separated Value (CSV) file to simulate a database. A dedicated Quick Response (QR) scanner enables rapid student identification and transactions, and encryption algorithms protect private information.

The prototype development successfully adhered to the \$100 budget constraint. A comprehensive financial plan has been developed for full-scale implementation, outlining projected costs and potential revenue models. These models include exploring an optional student fee for the service or strategic partnerships with on-campus businesses.

Evaluation of the digital student card system highlights its significant potential. The user interface design and data management components received particularly positive evaluations. Recommendations for future development include completing the login functionality, strengthening encryption protocols, implementing QR code generation capabilities, and seamlessly integrating the system with Queen's existing student card infrastructure.

Team E is confident in the viability of the digital student card system. Upon successful implementation at Queen's University, this innovation offers the potential for adoption by other universities seeking to modernize and streamline their campus experiences.

Table of Contents

1. Innovation Summary.....	1
1.1 Innovation Objectives	1
1.1.1 Stakeholder Identification	1
1.1.2 Determined Constraints.....	1
1.2 Background Information	2
1.2.1 Transaction Processing	2
1.2.2 Front-End Development	2
1.2.3 Back-End Development	3
1.2.4 Security Measures	3
1.3 Innovation Description	4
1.3.1 Transaction Processing	4
1.3.2 Application Front-End.....	5
1.3.3 Application Back-End	8
1.3.4 Security and Encryption.....	8
1.3.5 Environmental and Societal Costs and Benefits	9
1.3.6 Decommissioning Plans	9
1.4 Conclusions	10
2. Technical Information	10
2.1 Potential Innovations and Decision Making	10
2.1.1 Potential Innovations.....	10
2.1.2 Decision Making.....	11
2.2 Implementation	11
2.2.1 Transaction Processing	12
2.2.2 Application Front-End.....	12
2.2.3 Application Back-End	14
2.2.4 Security and Encryption.....	14
2.2.5 Environmental and Societal Costs and Benefits	15
2.2.6 Modelling and Analysis	15
2.3 Project Plan	15
2.4 Financial Analysis	16

2.5 Evaluation	17
References	19
Appendix	21

Table of Figures

Figure 1: Prototype system with QR code in wallet, QR scanner, and QR scanning application.	4
Figure 2: QR codes corresponding to user accounts in the application back-end.	5
Figure 3: A digital student card added to an Apple Wallet.	5
Figure 4: Flowchart of front-end operation.	6
Figure 5: Front-end UI prior to first QR code being scanned.	7
Figure 6: Front-end UI after first QR code is scanned.	7
Figure 7: Examples of error pop-up messages.	8
Figure 8: Showcase of CSV database structure.	8
Figure 9: Selection structure used for validating Flex input.	13
Figure 10: Mockup of UI with the login page on the top and the home page on the bottom.	22
Figure 11: Updated WBS.	23
Figure 12: Updated project development Gantt Chart.	24

Table of Tables

Table 1: Bill of materials for digital student card prototype.	16
Table 2: Additional bill of materials required for further project development.	16
Table 3: Evaluation matrix used for selecting a top innovation, based on financial, societal, technical, scalability, and unique qualities.	20
Table 4: List of locations for QR scanners.	24
Table 5: Project success evaluation rubric.	25
Table 6: Work distribution chart for document as well as project.	26

1. Innovation Summary

The proposed digital student card aims to revolutionize campus life. This innovative solution offers increased convenience, streamlined transactions, and a commitment to sustainability by reducing plastic waste. It will integrate with existing campus systems, allowing students to enjoy seamless meal plan usage, enhanced security, and real-time access to important campus information.

1.1 Innovation Objectives

For Queen's University students, the digital student card is an innovation to provide convenience and real-time campus information, help reduce plastic waste, and enhance both the physical security of the card and the security of a student's personal information. Incorporated as software into mobile devices, it shall allow students to use their Meal Swipes, Trade A Meals (TAMs), Flex Dollars, and Athletics and Recreation Center (ARC) membership without the need to carry a physical card. For identification purposes, the student's photo and name will be accessible through the computer interfaces at dining halls and other university locations. Enhanced security measures include the use of a phone password which would also aid in preventing malicious use, in conjunction with added cybersecurity methods Team E plans to integrate discussed later in Section 1.2.4 Security Measures. Team E plans to implement the digital student card alongside the physical card, being a convenient complement to the existing system.

1.1.1 Stakeholder Identification

Aside from students, additional stakeholders include the university, the food services team, and the ARC team. The proposed innovation contributes to Queen's University's sustainability goals as detailed in their Climate Action Plan by reducing plastic waste [1], and streamlining operational processes for the university and university staff, through the hastening of the card swipe process. All the while tracking important information, such as a student's TAM usage and real-time dining location capacity. Upon successful implementation, as outlined in Table 5 other universities may become involved to incorporate a similar system at their institutions.

1.1.2 Determined Constraints

Recognizing the need for a realistic approach within twelve weeks, the team carefully considered constraints related to the development of the innovation such as the complexity of creating the user interface, dealing with data storage, and securing partnerships with the university and catering companies. Hence, the scope of the innovation has been narrowed to a functional prototype, outlined in Section 1.3 Innovation Description. This, coupled with a formal business proposition presented to Queen's, is what the team is confident of completing in the period provided.

1.2 Background Information

The research conducted for this project covers key technological areas including transaction processing methods, front-end and back-end development, and data encryption and security. The information gathered was used and supplemented to inform and justify decisions made during development.

1.2.1 Transaction Processing

Transaction processing methods encompass the hardware and software that facilitate the exchange of payment and information between the student and the university. The primary focus of the digital student card is on contactless transactions, and a justification for this approach is warranted. Several current market trends support a shift toward digital solutions. Contactless payment solutions are experiencing growth across sectors including entertainment, retail, hospitality, and healthcare. The market value of mobile point-of-sale (mPOS) terminals is projected to reach \$70 billion by 2027, fueled in part by the growth of electronic commerce accelerated by the COVID-19 pandemic. Contactless payment offers greater speed and convenience compared to other payment methods [2]. These trends suggest that a digital card alternative would align with contemporary solutions and benefit from an established user base.

Two primary choices for contactless payment are Quick Response (QR) codes and Near Field Communication (NFC), each with respective benefits and drawbacks. NFC offers advantages in speed and built-in encryption, leading to its widespread use in payment acceptance. Additionally, NFC functions independently of internet connectivity. Conversely, QR codes are significantly less expensive and easier to implement. Their versatility allows them to store diverse information, including contact details and web links, beyond payment data. Factors such as available funding, accessible technology, and customer technology must be considered when selecting a solution [3]. Due to ease of setup and adaptability, QR codes were chosen for implementation in the final product.

Further research was necessary to select a QR scanner model suitable for project needs, with an emphasis on low cost and high compatibility to minimize expenditure and ensure ease of implementation. A Universal Serial Bus (USB) wired QR scanner from the brand Eyoyo was selected through Amazon based on its positive user reviews, affordability, and versatility. At \$39.99 Canadian Dollars (CAD), the scanner was significantly less expensive than comparable models with similar ratings, which averaged around \$60 CAD or higher. The model's ability to function without additional software was an additional benefit [4]. These factors contributed to its selection as a suitable and cost-effective solution for the project's front-end hardware.

1.2.2 Front-End Development

Front-end development encompasses the interactive hardware and software used by staff to conduct operations, such as checking student data or confirming Meal Swipe usage when interfacing with a digital student card. While several options were initially considered, the team decided to utilize a Java Graphical User Interface (GUI) built with the Swing toolkit. This choice was based on the team's experience with coding user interfaces in this programming language. Effective GUI design necessitates an understanding of various Java systems and their programming principles. The Swing toolkit provides

functions for creating, coding, and implementing components such as buttons, text fields, and images. The Grid Bag Layout system facilitates proper formatting and spacing of these components. Additionally, utilizing time and date packages allows for the incorporation of current time data into the system [5]. Research and implementation of functions for creating visually appealing "pill" buttons were conducted to enhance the final design's aesthetic [6]. These functions and tools enable seamless implementation of the front-end components.

While functionality is paramount, it is not the sole focus of front-end development. Consideration of additional factors is necessary to create an appealing and user-friendly experience. Design principles advocate for creating initial mockups before coding. Attention to details such as font selection, whitespace usage, and overall visual cohesion is important in GUI design. Maintaining a simple and minimalistic approach is often preferred. User feedback is crucial in the development of a quality GUI [7]. These considerations ensure a positive and seamless experience for staff interacting with the digital student card system.

1.2.3 Back-End Development

Back-end development refers to the underlying programming that connects the digital student card to the databases necessary for its functionality. A website's back-end typically comprises three main parts: a server, an application (app), and a database. The server processes requests and manages network resources. The app houses the code and logic that drive the website's functionality. The database is a structured collection of data used for information storage and retrieval. Structured Query Language (SQL) is a specialized programming language used to interact with databases, allowing for data queries and manipulation [8].

Due to project scope limitations preventing direct integration with Queen's servers and databases, a sample database was required to demonstrate prototype functionality. Java and Comma Separated Values (CSV) files were selected for creating a back end and mock database for testing purposes. Implementing robust authentication and authorization microservices is crucial for back-end security due to the sensitive nature of the data it manages. User authorization safeguards against unauthorized data access and modification. Security is discussed further in Section 1.2.4 Security Measures. Application Programming Interfaces (APIs) are necessary to facilitate communication between the database and the separate program app [9]. File input and output libraries provide functions for reading and writing data to files [5]. This research establishes the foundation for the back-end programming and implementation within the prototype.

1.2.4 Security Measures

As briefly mentioned in Section 1.2.3 Back-End Development, security is an important element of back-end coding necessary to protect information going through the system. Key practices useful for protecting sensitive data and services include authenticating and authorizing access, implementing stringent access control measures, encrypting requests and responses, validating data, and conducting regular risk assessments. Documentation, security testing, and the careful management of API keys are also effective strategies [10]. Protecting the back end of the prototype would mean encrypting the CSV

file so that only those with authorized access could access the information within. This can be done using the Java crypto libraries which provide functions to encrypt and decrypt information using user set keys. With built functions the process of encrypting and decrypting information from a file is made easy [11]. These are the considerations made to securing the data used by the prototype.

1.3 Innovation Description

The delivered Minimum Viable Product (MVP) is a program which can scan QR codes and securely interact with a back-end database to verify identity for the ARC and spend Meal Swipes, Flex Dollars, and TAMs. Discussed below are the general functions of software and hardware components of the product, the societal costs and benefits provided, and the plans in case of decommissioning. The prototype is showcased in Figure 1 below.

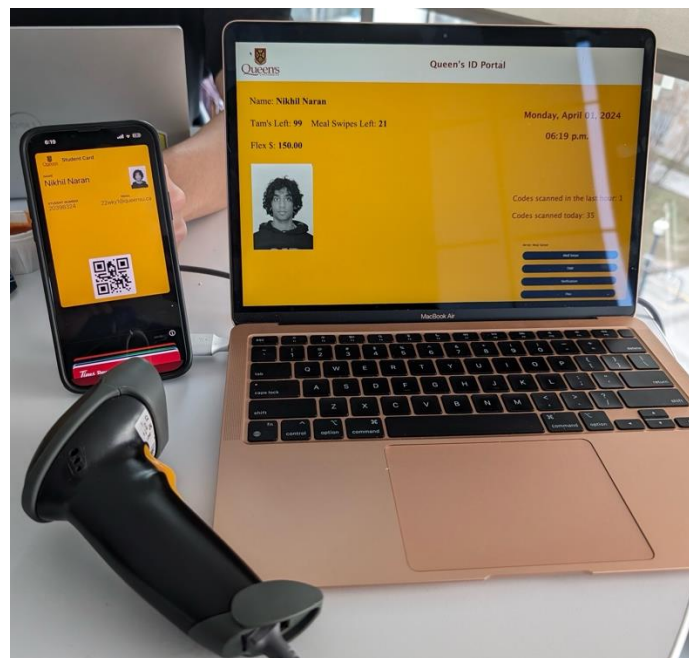


Figure 1: Prototype system with QR code in wallet, QR scanner, and QR scanning application.

1.3.1 Transaction Processing

The prototype interacts with digital student cards using a QR scanner. As detailed in Section 1.2.1 Transaction Processing the selected scanner is the cost-effective and versatile Eyoyo Wired USB 2D Barcode Scanner. Connected to the computer running the Java program, the scanner transmits scanned codes for processing based on the selected mode, which is discussed below. The scanner offers two operational modes: one actively scanning for codes and another that activates upon pressing a trigger.

A set of QR codes as seen in Figure 2 were generated for the prototype through a built in iPhone QR generating software [12], each linked to a student account within the mockup database (described in Section 1.3.3 Application Back-End). The QR codes created were added an Apple Wallet shown in Figure 3 using a third-party app called Pass2U. Although only an Apple Wallet card was created for the prototype the app can generate cards for Google Wallets as well [13].

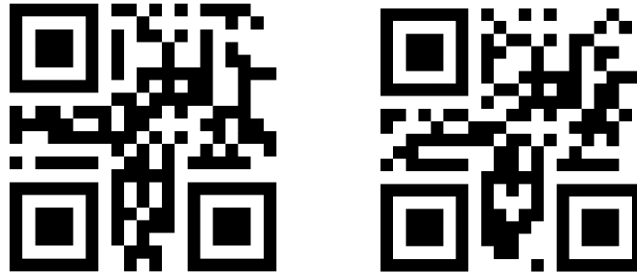


Figure 2: QR codes corresponding to user accounts in the application back-end.

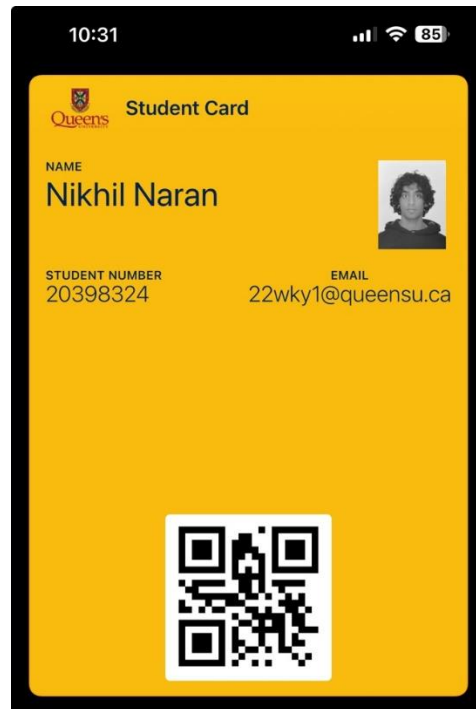


Figure 3: A digital student card added to an Apple Wallet.

1.3.2 Application Front-End

The Queen's University staff will interact with the product through a designed app. Figure 4 outlines the workflow through which staff will use said app, and Figure 5 and Figure 6 illustrate its interface. Additionally, Figure 9 in Section 1.3.2 Application Front-End expands on the verification process for determining if the input Flex amount is valid. Initially, staff select an operational mode: "Meal Swipe," "TAM," "Flex," or "Verification." Upon scanning a QR code, relevant student information (name, number of TAMs, Meal Swipe balance, Flex Dollar balance, and photo) are displayed. The selected mode determines the subsequent action, allowing for immediate processing and facilitating faster transactions, especially during peak hours.

The front-end incorporates pop-ups and error handling mechanisms (Figure 7). In "Flex" mode, a text field prompts the user to enter the Flex Dollar amount. The program will only proceed with the transaction if a valid, positive numerical value is entered. In the event of insufficient TAMs, Meal Swipes,

or Flex Dollars, an error message will be displayed, preventing any erroneous deductions from the student account. Additional features include real-time display of date and time, along with counters for codes scanned within the hour and day, providing staff with operational insights.

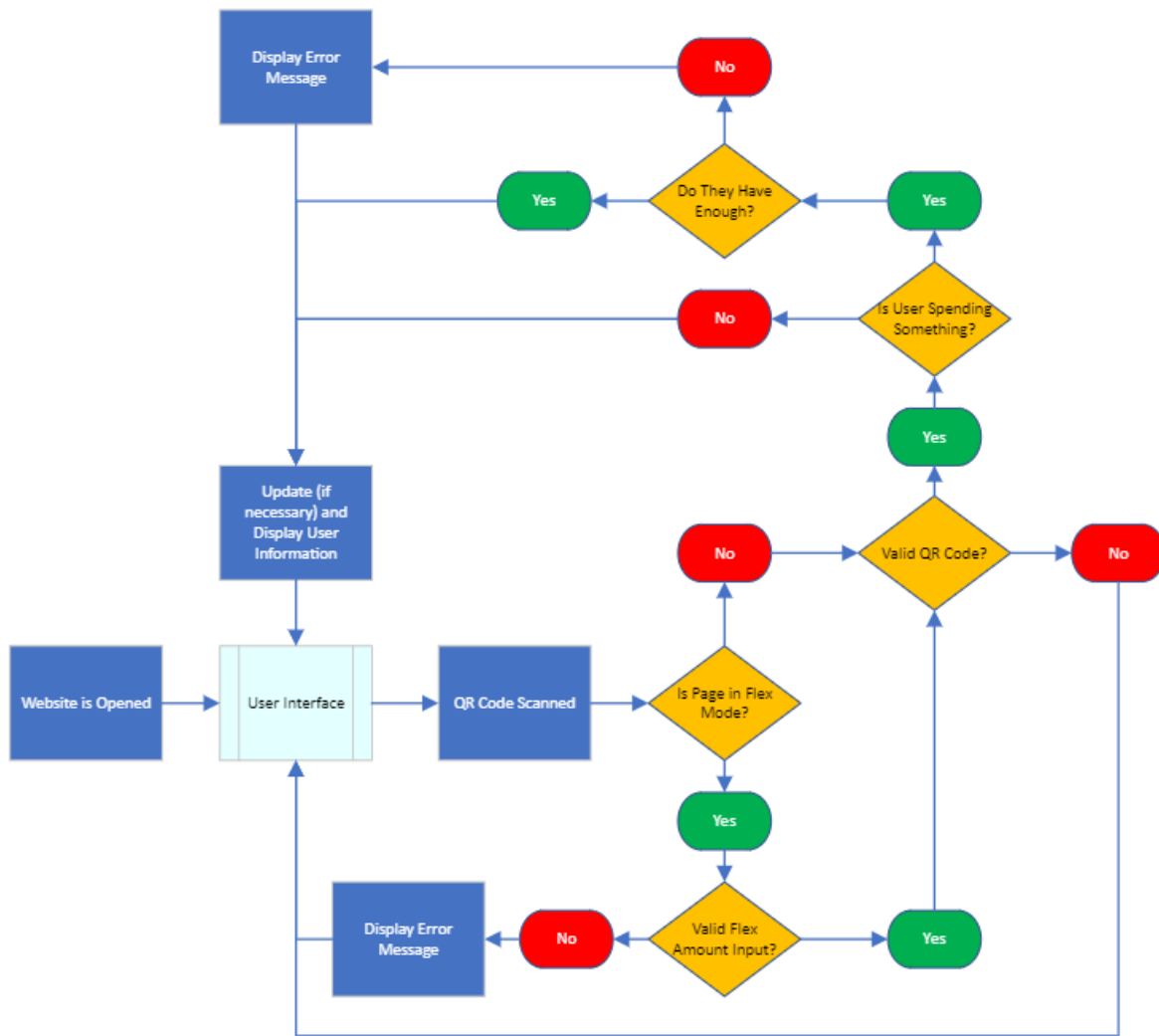


Figure 4: Flowchart of front-end operation.

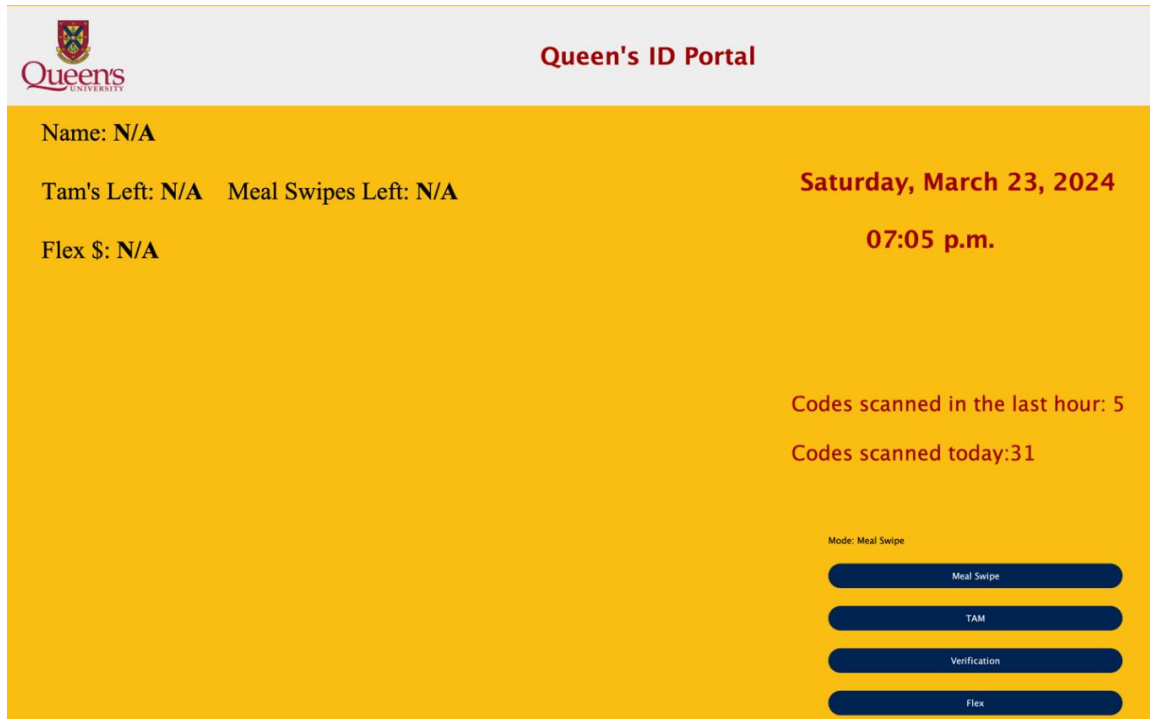


Figure 5: Front-end UI prior to first QR code being scanned.



Figure 6: Front-end UI after first QR code is scanned.

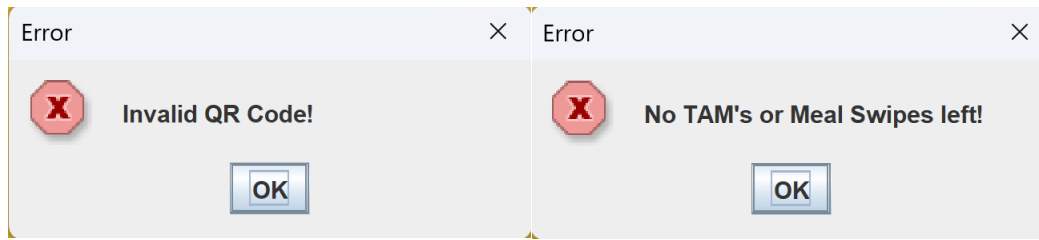


Figure 7: Examples of error pop-up messages.

1.3.3 Application Back-End

The app's back end facilitates communication with the CSV file database which stores student information. The formatting of this mockup database is seen in Figure 8 below and functions just like a real database using table like structures to organize data and make it easy to access. The mockup demonstrates input and output testing and provides the foundation for implementing a full-scale database. As discussed in Section 1.2.1 Transaction Processing each QR code references a specific location within the database where the corresponding student's information is held. Upon scanning a QR code, the reference information is validated against the list of references within the database to confirm user existence. If verification is successful, the student's information is retrieved and transmitted to the front-end for display and potential updates. Updated values are then returned to the database for storage.

```

1      ,N/A,Empty.png,N/A,N/A,N/A
2      code, name, image, tams left, meals swipes left
3      064541310790,Nikhil Naran,nik.jpeg,99,20,150.00
4      030772030868,Fedor Kim,test.jpg,108,20,150.00

```

Figure 8: Showcase of CSV database structure.

1.3.4 Security and Encryption

The security component is paramount and involves protecting end-user data, as outlined in Section 1.2.4 Security Measures. Upon program launch the CSV file is encrypted and can only be decrypted by authorized individuals possessing the secret key. This setup ensures data is protected even if a QR code is scanned without authorization. Access to personal user information is restricted to those with the correct key, which safeguards data from unauthorized access. Without the correct key, the database remains inaccessible even with a reference provided by a QR code, preventing data theft.

On the end user side security is achieved through built in mobile device encryption and safety measures. To access a digital student card the user must enter their mobile device password or use face recognition software which prevents others from being able to use their card. As a photo of the user is included on every card and is displayed upon scanning a QR code, only the card holders are able to use their card.

1.3.5 Environmental and Societal Costs and Benefits

The implementation of a digital student card system aligns with the team's value proposition to improve student experience and streamline campus operations. Students opting to use the digital card will worry less about losing their card or fumbling for it at entrances to dining halls as access will be instant through their smartphones. Contactless payments speed up transactions, a crucial benefit during peak periods like move-in week or busier hours of operation. Furthermore, the contactless element will help curb the spread of disease during Orientation Week and flu season. Robust security measures like encryption and two-factor authentication will protect student data and reduce the risk of fraudulent use. Team E also acknowledges a crucial societal cost: the potential barrier for students from low-income backgrounds who might not have smartphones. To ensure inclusivity, the digital student card system will be a complementary option alongside the existing physical card system. This allows students who lack smartphones to continue using their physical cards without disruption. Furthermore, maintaining physical cards alongside the new digital system addresses concerns about potential cheating during exam season. With the digital system, there could be an increased risk of students using their smartphones to access unauthorized information during exams. Requiring physical student cards in these situations helps to uphold academic integrity.

Environmentally, one of the most significant benefits is the decreased need for physical card production. With a digital system, the frequency of lost or damaged cards requiring replacement will dramatically decrease. This directly translates to less plastic waste and a reduced carbon footprint associated with card manufacturing. Team E is committed to investigating even more sustainable options for the future. This could include partnering with a card provider that uses recycled content or exploring fully biodegradable physical card options to replace the continuing physical cards.

1.3.6 Decommissioning Plans

While deemed improbable, the possibility exists that Queen's University may encounter an insurmountable obstacle hindering the implementation of the proposed digital student card system. This could stem from various factors, with the most significant being technical or security failures. The emphasis placed on success in these areas is reflected within Section 2.5 Evaluation) which aims to identify and mitigate such risks. However, should legal, financial, or maintenance challenges arise that impede project completion, a pre-defined decommissioning plan will be activated to ensure the continued functionality of the student card system.

Since the digital card is not intended to fully replace its physical counterpart, reverting to the existing system would be a straightforward process. The QR scanners implemented for digital card interactions could be repurposed or uninstalled. As for sensitive student information, secure deletion would not be required as the aim of the product would be to integrate with existing databases used for student cards. The only consideration would be to close any connections between the digital student cards and database. As the framework for the physical card system would remain intact, it would seamlessly resume its previous functions. Apart from potentially re-adjusting the scale of student card replacements, no additional decommissioning costs are anticipated. Ultimately, the project strives to offer a convenient alternative while preserving the existing system for those who prefer it.

1.4 Conclusions

The proposed digital student card system demonstrates the potential to enhance the student experience at Queen's University. Designed to complement the existing physical card system, this innovation offers students increased convenience and a streamlined method for accessing campus services. By using their smartphones as a digital card, students can effortlessly make meal plan purchases, redeem TAMs, utilize Flex dollars, and access the ARC. This transition also aligns with the university's sustainability goals by significantly reducing plastic waste.

Importantly, the project prioritizes security. Encryption protocols and other safeguards protect student information. Careful consideration has been given to ensuring inclusivity for students without smartphones, as the physical cards will remain in use alongside the new digital option. In the unlikely event of technical or security obstacles, a detailed decommissioning plan ensures a smooth return to the existing physical card system. Overall, the digital student card initiative promises to modernize campus life, offering benefits for students, the university, and the environment.

2. Technical Information

Technical information related to the project is included in this section of the document. Outlined below are the steps used to select an innovation, the process of and considerations made during the innovation's implementation, a review of the final project plan, a financial analysis of the product, and the evaluation of the prototype.

2.1 Potential Innovations and Decision Making

Several ideas were brainstormed to innovate upon existing products and processes. The team considered five main innovation ideas, including a redesigned OnQ, smart mug, note-sharing app, digital student card, and convenience app. Selecting an innovation to further develop required evaluation of the options which is discussed further in this section.

2.1.1 Potential Innovations

The redesigned OnQ would have consisted of reimagining the existing OnQ platform to provide students' schedules and assignments online and to simplify deadline management and communication. This would have benefited students, faculty, and Queen's University through improved organization and potentially higher-grade point averages. Next, the smart mug's objective was to integrate a gimbal, heater, and stirrer for a spill-proof and temperature-controlled drink, targeting students and adults. Third was a note-sharing app idea intended for students to share their class notes and then have those notes verified by professors or teaching assistants, helping others catch up and boosting success. The digital student card would be a digital form of the existing student card system, introduced to help fix physical student card issues. The convenience app is a service which would deliver essential items like toilet paper, plastic cups, and other daily necessities directly to students' doors.

2.1.2 Decision Making

Brainstormed ideas underwent a rigorous evaluation process, prioritizing societal need, target market, financial viability, and technical feasibility. The aim was to select an innovation that directly addressed a pertinent issue while remaining achievable within the project's constraints. These constraints included a \$100 budget for materials and expenses, a 12-week timeline, and the requirement for seamless integration with Queen's existing systems. There were five innovation ideas considered, including the digital student card, a redesigned OnQ, a smart mug, a note-sharing app, and a convenience app. Ultimately, among the conceived ideas, the digital student card was selected due to its potential for significant positive impact on Queen's students, coupled with its realistic development trajectory. The evaluation matrix (Appendix, Table) informed this decision and the scores provided are justified below.

- **Digital Student Card:** Addresses a prevalent issue of lost or misused cards for Queen's students. It is financially viable, with limited costs beyond initial development. Importantly, it aligns well with the project's technical constraints, primarily requiring integration with existing systems. While scalability to other institutions might be limited and similar technologies exist, it offers a strong solution tailored to Queen's unique needs.
- **Redesigned OnQ:** This project offers potential financial viability and addresses a clear societal need to improve student organization and academic performance. However, the extensive redesign of a complex, existing system presents significant technical challenges within the project's timeframe and budget. While scalable, most educational institutions have their own customized course management systems, reducing its uniqueness.
- **Smart Mug:** While offering a distinctive combination of features, the smart mug's development costs, niche market appeal, and complex integration of heating, mixing, and stabilization technologies present financial and technical hurdles within the project's scope.
- **Note-Sharing App:** This app addresses a genuine societal need and offers potential scalability. However, the costs associated with developing and maintaining the app, particularly the incorporation of machine learning features, pose a financial challenge. While the core note-sharing functionality is feasible, the advanced machine learning aspects create potential technical complexities with limited resources.
- **Convenience App:** This idea has strong potential for financial success and scalability. However, it primarily addresses a desire for convenience rather than a core societal need. Additionally, developing a full-scale, functional app within the timeframe is technically ambitious. The concept lacks uniqueness, as similar convenience platforms already exist.

2.2 Implementation

This section of the document aims to provide an in-depth description of how the prototype functionalities outlined in Section 1.3 (Innovation Description) were coded and implemented. Furthermore, this part of the document discusses the modelling and environmental and societal considerations made during its development.

In a general statement concerning the implementation of the project, there were several instances where the team utilized the assistance of ChatGPT to bridge minor informational gaps and further enhance the overall quality of the final product. Such an approach was chosen due to the team's low technical experience in areas of coding and was only used to supplement the team's work, when necessary, not replace it. This Artificial Intelligence (AI) aide was only used to help with coding the prototype, not in the writing of this report, and the team firmly believes that this falls in line with the expectations set out in the course syllabus. Any instances where ChatGPT was utilized are described as they come up below.

2.2.1 Transaction Processing

Implementing the QR scanner required a simple connection to the computer running the Java program. As a Human Interface Device (HID), the QR scanner decodes a scanned code into text, followed by a newline character, simulating keyboard input. To receive codes, an action listener was added to the GUI's main panel. This listener captures text entered after a newline character. Initially, an error occurred when adding buttons and a text field for mode selection, which shifted focus away from the panel and prevented proper QR code scanning. This was resolved by coding each button with a command to return focus to the program's main panel.

2.2.2 Application Front-End

As outlined in Section 1.2.2 Front-End Development), the prototype's Graphical User Interface (GUI) was built using Java Swing. This toolkit provides the components and visual interactions within the app. Java Swing was chosen due to its lightweight nature, cross-platform compatibility, and ease of development. The Grid Bag Layout system forms the framework of the app, allowing dynamic placement of objects and components. The app comprises several distinct components: a header panel, button panel, date and time panel, and an information panel. These panels are integrated through the Grid Bag Layout.

Due to the complex configuration required by the layout system, ChatGPT assisted with generating the code to achieve the desired spacing and formatting of components. The Queen's logo was imported as a separate image, placed in the top left corner. Coloring aligns with the Queen's color palette using Red Green Blue (RGB) commands. Text was made bold for readability and to improve the overall program aesthetic. Custom Java classes were used to create visually prominent "pill-shaped" buttons for user interaction.

Going into further depth regarding the programming of components, each button is implemented with an action listener which runs a set of code when the button is pressed. The general function of each button is to update the text area which describes the selected mode. The updated text area is what the back end will then look at to determine the selected mode and next operation after a QR code is scanned. The Flex button contains additional code as it runs a command to make the Flex text field visible (every other button makes this text field invisible to the user). The Flex text field allows for the input of Flex and is programmed with a selection structure that only allows for a QR code to be scanned after a valid amount of Flex is input. The selection structure (Figure 9) checks for the following things: there is an input in the Flex field, the Flex amount contains a minimum of no decimals and a maximum

of one, the Flex amount is made up of all digits, and the input Flex amount is a positive value. Only after a valid Flex amount is entered will this text area run the code to return focus back to the main panel of the program which as mentioned in Section 2.2.1 Transaction Processing will allow for a QR code to be scanned.

The current date and time display involves additional programming. These values are retrieved from the device running the program using dedicated local date and time libraries. A timer function executes code every second, updating the date and time text area. The code retrieves the local date and time, formats them according to an imported formatter, and outputs the formatted text. ChatGPT assisted with generating the necessary commands for working with Hyper Text Markup Language (HTML) date and time content.

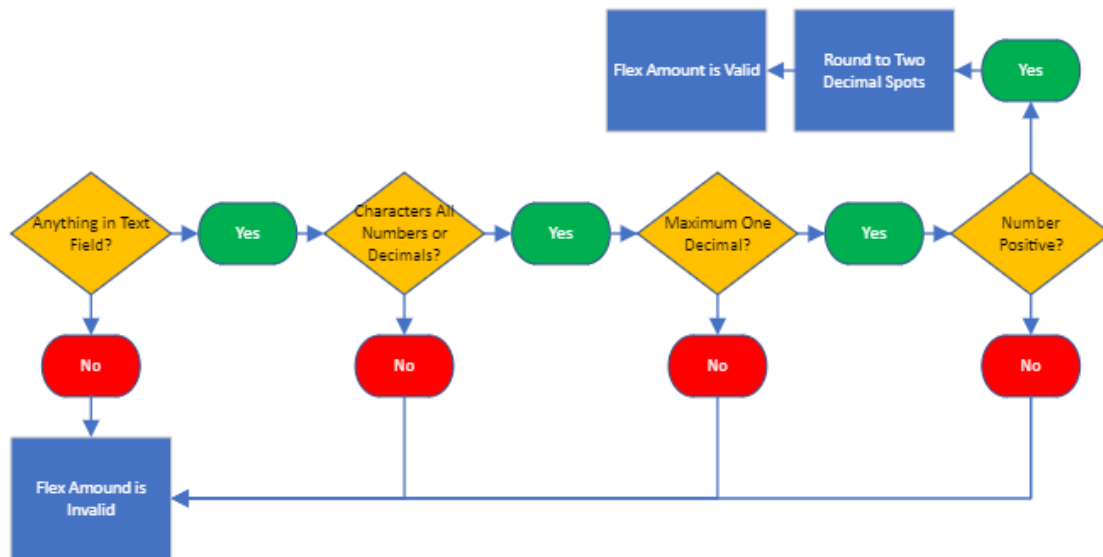


Figure 9: Selection structure used for validating Flex input.

The "codes scanned" feature also utilizes timers. However, instead of a dedicated timer, this functionality leverages a counter that tracks the app's total runtime. Each time a new code is scanned, the program records the time of the scan and increments the "codes scanned in the last hour" and "codes scanned in the last day" counters. Corresponding text areas on the GUI are also updated. Time values are stored in a linked list. The program periodically checks if any stored time values exceed one hour or one day relative to the current runtime. If scans are older than the specified thresholds, they are removed from the list, and the relevant text areas are updated. Additionally, upon window closing, the total number of codes scanned is saved to a data file for persistent storage.

Finally, the front-end dynamically updates text areas to display cardholder information as QR codes are scanned. Upon initial launch, a single-use code segment retrieves a test case from the CSV file and displays it, as seen in Figure 4. This ensures consistent page formatting and early identification of file output issues. Subsequent valid QR code scans trigger the front-end to display relevant user information pulled from the back end, which is discussed in detail next. ChatGPT was used to assist with generating commands for converting strings into HTML and vice versa.

2.2.3 Application Back-End

The app's back end primarily focuses on reading and writing data to a CSV file. Data reading is accomplished using a combination of Buffer Reader and File Reader, built-in Java functions. Data writing utilizes Print Writer and File Writer, also built-in Java functions. Once a QR code is scanned and entered, the system verifies its existence within the CSV file using a purpose-built function. An error message is displayed if the code is not found. The system also checks if the scanned code differs from the previous code to prevent accidental double-scans, especially with a passive scanning mode.

After successful code validation, relevant user information is sent to the front-end for display. The system then uses a selection structure to determine the currently selected mode (Meal Swipe, TAM, Flex, or Verification). Based on the mode, the system verifies sufficient remaining Flex, Meal Swipes, or TAMs, updates the values accordingly, and writes these changes back to the CSV file. The CSV file serves as a database, storing essential information for each user, including name, remaining Meal Swipes, TAMs, and Flex. It also stores references to user images located elsewhere in the back end.

2.2.4 Security and Encryption

The development of a security solution involved the usage of pre-built functions for the “PBEWithMD5AndDES” encryption algorithm [11], as briefly mentioned in Section 1.2.4 Security Measures) above. Although mostly finished, these encryption and decryption functions had to be modified to properly implement into the project. The “PBEWithMD5AndDES” algorithm serves as an advanced mechanism for protecting digital data, with an emphasis on security of the QR code information. Initially, the chosen password undergoes a transformation using the Message-Digest algorithm 5 (MD5), which encrypts data in a unique manner. Then, the Data Encryption Standard (DES) encryption standard is applied to further obfuscate of the information, ensuring that access is confined only to individuals possessing the correct password. To augment security, the introduction of a “salt” — a random piece of data — ensures that even identical passwords will yield unique encryption outcomes. The security process is further enhanced by repeating the mixing action 100 times, which reduces chances of unauthorized access.

The decryption process is the reverse of the encryption. It begins identifying and retrieving the “salt” that was integrated during the encryption phase from the encrypted file. Utilizing the original password along with the “salt,” the secret key is reconstructed. The original password, in conjunction with this salt, is used to regenerate the secret key, mirroring the encryption's initial key generation. This preparation aligns the “PBEWithMD5AndDES” algorithm for decryption, setting it to revert the encrypted data back to its original state. Access is restricted to individuals possessing the correct password, thus making data secure from unauthorized access.

2.2.5 Environmental and Societal Costs and Benefits

The development of the digital student card system prioritized sustainability and usability. The choice of a software-focused solution minimized resource use and energy consumption during creation. While the primary benefits of the innovation will come from reduced plastic card production, several societal factors are considered as well. One such factor was the QR code scanning process. To ensure faster

verification and a more seamless user experience, the team opted to integrate a dedicated QR code scanner rather than relying on phone cameras. Security concerns were also crucial; data encryption, as detailed in Section 2.2.4 Security and Encryption, was implemented to safeguard student information before and after using it. Finally, recognizing the importance of user adoption, the team prioritized creating an intuitive and aesthetically pleasing user interface. This user-centric approach was aimed to enhance the overall user experience and encourage widespread acceptance of the new digital card system.

2.2.6 Modelling and Analysis

The modelling completed during development of the project involved creating a mockup of the front-end to identify required as well as desired features. Shown in Figure 10 in the Appendix, the nonfunctional mockup of the front-end was used as a basis for coding and implementing the features seen in the final product. Several innovative ideas included in the mockup were carried over to the prototype including the codes scanned in the past day and hour as well as the display of the date and time. A feature that was expanded upon was the buttons for spending TAMs and Meal Swipes. In the final product additional buttons for spending Flex and viewing student information were coded and implemented to better match the objectives outlined in Section 1.1 Innovation Summary. Finally, due to time constraints, the login page functionality was not fully developed for the final prototype. To prioritize features more central to the core concept, the decision was made to focus on resource allocation towards other aspects of development. This limitation and its rationale are discussed further in Section 2.5 Evaluation.

2.3 Project Plan

The updated Work Breakdown Structure (WBS) and related Gantt chart can be found in Figure 11 and Figure 12 (in the Appendix) respectively. Special consideration was taken when designing the project plan to account for the possibility of project delays brought by many possible issues the team would occur. To mitigate the risks of possible issues, the team used overestimation of time required to complete each project task. 3-day buffers were added to the expected duration for each main part of the project i.e. sections 2, 3, 4 of the WBS. This scheduling flexibility allowed for mobility of the team's task deadlines, ensuring key project milestones are met under manageable stress levels. Unfortunately, the team was not able to implement dynamic QR code generation in the finished prototype, and this area was pushed to further development as it was not strictly necessary for the MVP. Instead, there was an emphasis on the usability of the UI and the efficiency of the encryption of the data. In retrospect, more time should have been allocated for the security and encryption of the data. The encryption task took the team much longer than expected. Using GitHub repositories, the rest of the team was able to direct their attention to the front-end of the project, while the security function was worked on simultaneously. However, the delay slightly pushed back the testing and finalization of the product. Overall, the original project plan was done well, giving time for potential risks, but changes were needed throughout the project as described above.

2.4 Financial Analysis

The development of the digital student card prototype managed to remain within the \$100 budget set by the APSC 103 course requirements. The project budget included the purchase of the QR code scanner and the utilization of software development tools such as GitHub and IntelliJ Integrated Development Environment (IDE). It should be noted that the free educational licenses for GitHub and IntelliJ can only be used for non-commercial purposes. Used resources are outlined in Table below.

Table 1: Bill of materials for digital student card prototype.

Item	Description	Quantity	Pricing CAD
QR code scanner	Hardware part	1	\$39.99 [4]
GitHub	Software Development tool for collaborative work	5 (for each team member)	Free for students
IntelliJ IDE	Software Development tool for Java	5 (for each team member)	Free for students

Moving beyond the course and making the transition from a prototype to a fully operational Digital Student Card system incurs additional costs. These include system integration into the existing Queen's University student card database, labor costs, and further project development aimed at enhancing user accessibility and experience.

Regarding the costs associated with integration, the project could use Microsoft Azure cloud services for the web app [14], a JetBrains Tools package for software development and design [15], a GitHub repository for collaborative work [16], and an estimated additional 26 QR code scanners for the Queen's University campus. The QR scanner estimate includes food spaces as well as the ARC and a list of locations is provided in Table 4 (in Appendix). The minimum wage for a Software Developer/Programmer, based on statistics from the Government of Ontario [17], is estimated at 28.44 CAD per hour. The outlined expenses are summarized in a bill of materials seen in Table 2. Overall, the approximate costs for the integration of a Digital Student Card system at Queen's University, based on Table 2 and excluding the salary of the labor force, amount to \$8,704.07 for the first year of operation.

Table 2: Additional bill of materials required for further project development.

Items	Description	Quantity	Pricing CAD
QR code scanner	Hardware part	26	\$39.99 / unit [4]
GitHub Repository	Software Development Tool for collaborative work	-	\$19.25 / month [16]
JetBrains Tools Pack	Software Development Tool (IDE package)	-	\$779.00 / year [15]
Microsoft Azure cloud server D4 v2	Cloud Server, 4 Cores, 14 GB RAM, 200 GB storage	-	\$564.5203 / month [14]

For financing, the team proposes adding the digital student card as an optional yearly ancillary fee provided when paying tuition. This would allow students to opt in for a digital student card at an extra charge of \$1 CAD. Given an enrolment of 28,000 students in 2022 this fee would be sufficient to finance the project if around a quarter of the students apply [18]. Generally, this fee is designed to ensure affordability for all students at Queen's University while generating revenue that would be used for further project development and improvement. This would ensure the project's sustainability without imposing an additional financial burden on the university.

Targeted advertising is also an option, but the team acknowledges the importance of maintaining a positive user experience. Further investigation is required to determine the most suitable advertising formats and partners. The team is exploring alternative revenue models that do not rely heavily on advertising, such as partnerships with on or off campus businesses.

2.5 Evaluation

The team employed multiple methods to evaluate project success. A primary tool was the evaluation matrix (Appendix, Table 5), which assessed the new student card system against five criteria: core integration functionality, user interface design, data encryption, efficient data management, and QR code generation. These criteria were rated on a scale from "Needs Improvement" (1) to "Excellent" (4), with a maximum possible score of 20. To ensure an optimized user experience, a minimum total score of 12 was established, with no individual category scoring below 2. Due to time constraints and a redefinition of the project scope after Phase 3, the team focused on developing the core digital student card platform, postponing QR code generation for future iterations. Testing was also incorporated through food service staff, who evaluated the system's functionality and usability, providing valuable insights for improvement.

The final prototype earned a total score of 14 in the matrix. Core functionality (score of 3) demonstrated basic operations, though elements like the login page remain incomplete. The user interface (score of 4) received positive feedback, aligning with the project's goals for an intuitive and visually appealing design as seen in Figure 6. Data encryption (score of 3) is implemented, but potential vulnerabilities highlight the need for even more robust security measures in future development. Efficient data management (score of 4) proved successful, with seamless integration between the front-end and back-end of the system. Overall, the evaluation validates the system's potential to fulfill our value proposition and provides. It offers a strong foundation for further development, with clear guidance for refinements to enhance convenience, sustainability, and security for Queen's students.

References

- [1] Queen's University, "Climate Action Plan," Queen's University , January 2016. [Online]. Available: <https://www.queensu.ca/sustainable/about/climate-action-plan>. [Accessed 16 Debruary 2024].
- [2] Global Market Insights, "5 Major Trends Transforming mPOS Terminals Market Outlook Over 2021-2027," Global Trade Magazine, 29 September 2021. [Online]. Available: <https://www.globaltrademag.com/5-major-trends-transforming-mpos-terminals-market-outlook-over-2021-2027/>. [Accessed 27 January 2024].
- [3] Utilities One, "NFC vs QR Codes Choosing the Right Wireless Standard for Contactless Payments," Utilities One, 7 August 2023. [Online]. Available: <https://utilitiesone.com/nfc-vs-qr-codes-choosing-the-right-wireless-standard-for-contactless-payments>. [Accessed 27 January 2024].
- [4] Amazon, "Eyoyo Wired USB 2D Barcode Scanner QR PDF417 Data Matrix 1D Bar Code Scanner Handheld Barcode Reader with USB Cable for Mobile Payment, Convenience Store, Supermarket, Warehouse," Amazon, 2 July 2020. [Online]. Available: <https://www.amazon.ca/Eyoyo-Handheld-Convenience-Supermarket-Warehouse/dp/B088QV215Y>. [Accessed 2 March 2024].
- [5] R. Gallardo, S. Hommel, S. Kannan, J. Gordon and S. B. Zakhour, The Java Tutorial: A Short Course on the Basics (6th Edition), Boston: Addison-Wesley Professional, 2014.
- [6] VonC, "Rounded Swing JButton using Java," Stack Overflow, 10 November 2023. [Online]. Available: <https://stackoverflow.com/questions/423950/rounded-swing-jbutton-using-java>. [Accessed 10 March 2024].
- [7] B. Grezeszak, "Top 10 Front-end Design Principles for Developers," Toptal, [Online]. Available: <https://www.toptal.com/front-end/front-end-design-principles>. [Accessed 26 January 2024].
- [8] V. S, K. P Hathwar, R. Ravishankar, N. BN, H. R and A. HP, "Back-End Web-Application Development and the Role of an Admin," *IJARCCCE*, p. 60, 2017.
- [9] S. Sharma, "Backend Software Architecture Checklist: How to Build a Product from Scratch," freeCodeCamp, 23 April 2020. [Online]. Available: <https://www.freecodecamp.org/news/have-an-idea-want-to-build-a-product-from-scratch-heres-a-checklist-of-things-you-should-go-through-in-your-backend-software-architecture/>. [Accessed 26 January 2024].
- [10] M. Cobb , "10 API Security Guidelines and Best Practices," 2023. [Online]. Available: <https://www.techtarget.com/searchapparchitecture/tip/10-API-security-guidelines-and-best-practices>. [Accessed 26 January 2024].

- [11] P. White, "File encryption / decryption retaining line feeds," Coderanch, 2012. [Online]. Available: <https://coderanch.com/t/562936/java/File-encryption-decryption-retaining-line>. [Accessed 12 March 2024].
- [12] myakSTEM, "QR codes easily and securely," Apple, 17 August 2023. [Online]. Available: <https://education.apple.com/resource/250011714>. [Accessed 1 April 2024].
- [13] Pass2U, "Digitize Cards And Tickets to Apple Wallet / Google Wallet," Pass2U, [Online]. Available: <https://www.pass2u.net/>. [Accessed 26 March 2024].
- [14] Microsoft Azure, "Microsoft Azure Cloud Server Pricing," Microsoft Azure Cloud Server, February 2024. [Online]. Available: https://azure.microsoft.com/en-ca/pricing/details/cloud-services/?ef_id=_k_CjwKCAjwkuqvBhAQEiwA65XxQBxLve2F4VhQk5hbcS6jDVxXDxcE4vEfjoO9BeClbpv6oLnV97mPvBoCzX4QAvD_BwE_k_&OCID=AIDcmmqz3gd78m_SEM__k_CjwKCAjwkuqvBhAQEiwA65XxQBxLve2F4VhQk5hbcS6jDVxXDxcE4vEfjo. [Accessed 20 March 2024].
- [15] JetBrains, "Jetbrains Subscription Options and Pricing," JetBrains Software Products Pack, March 2024. [Online]. Available: <https://www.jetbrains.com/store/#commercial?billing=yearly>. [Accessed 20 March 2024].
- [16] GitHub, "GitHub Subscription Pricing," GitHub, March 2023. [Online]. Available: <https://github.com/pricing>. [Accessed 20 March 2024].
- [17] Government of Ontario, "Wages in Ontario," Government of Ontario, 23 01 2024. [Online]. Available: <https://www.jobbank.gc.ca/wagereport/location/on>. [Accessed 20 March 2024].
- [18] Queen's University, "2022-23 Enrolment Report," Queen's University Office of the Provost and Vice-Principal Academic, Kingston, 2022.

Appendix

Table 3: Evaluation matrix used for selecting a top innovation, based on financial, societal, technical, scalability, and unique qualities.

Criteria	Weight	Digital Student Card	OnQ Redesigned	Smart Mug	Note Sharing Application	Convenience Store
<i>Financial Viability</i>	20	18/20	10/20	6/20	11/20	19/20
<i>Social Need</i>	30	27/30	29/30	16/30	26/30	12/30
<i>Technical Feasibility</i>	30	24/30	16/30	18/30	15/30	12/30
<i>Scalability</i>	10	6/10	8/10	6/10	9/10	9/10
<i>Uniqueness</i>	20	10/20	11/20	18/20	14/20	8/20
Total	100	76	74	64	75	60

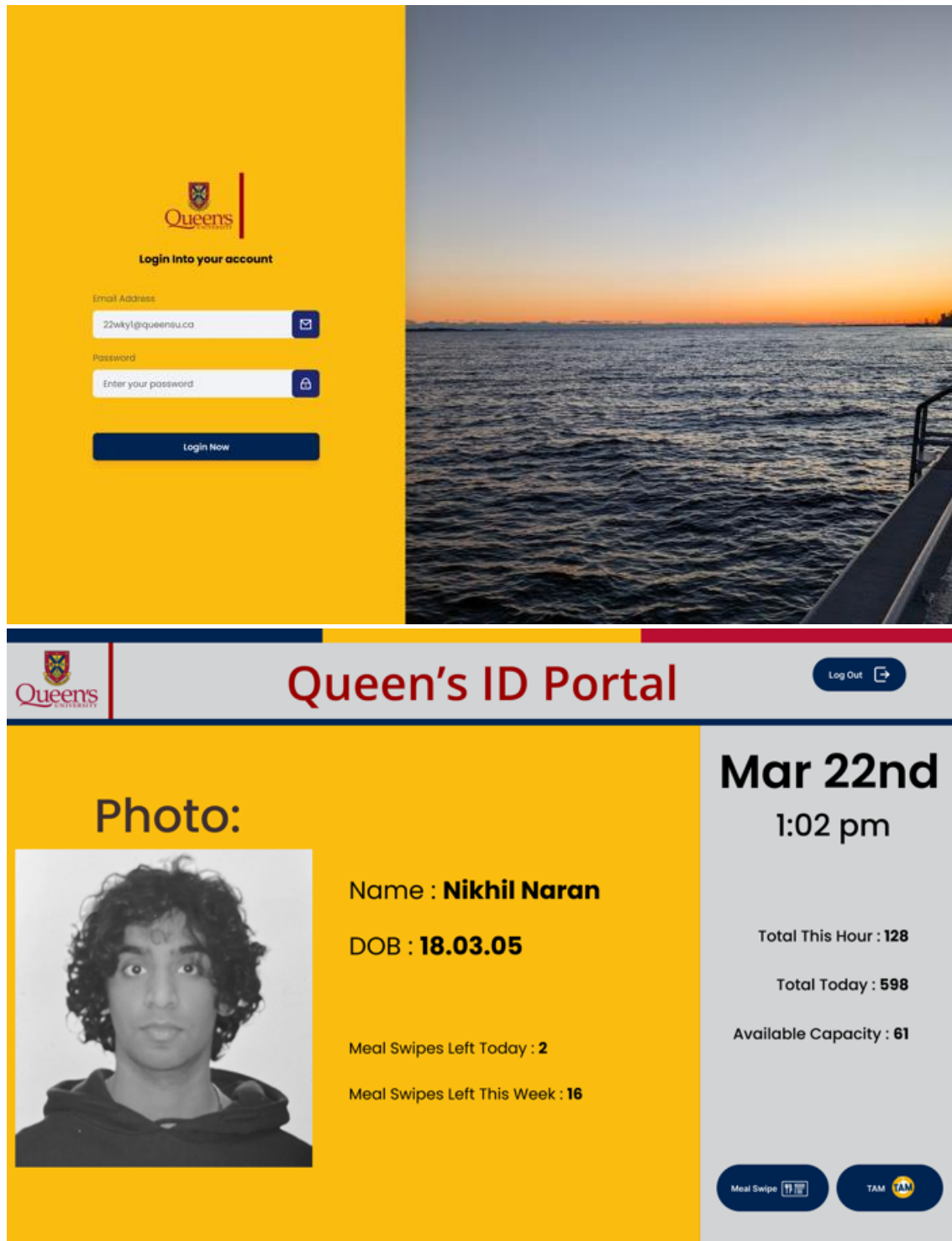


Figure 10: Mockup of UI with the login page on the top and the home page on the bottom.



Figure 11: Updated WBS.

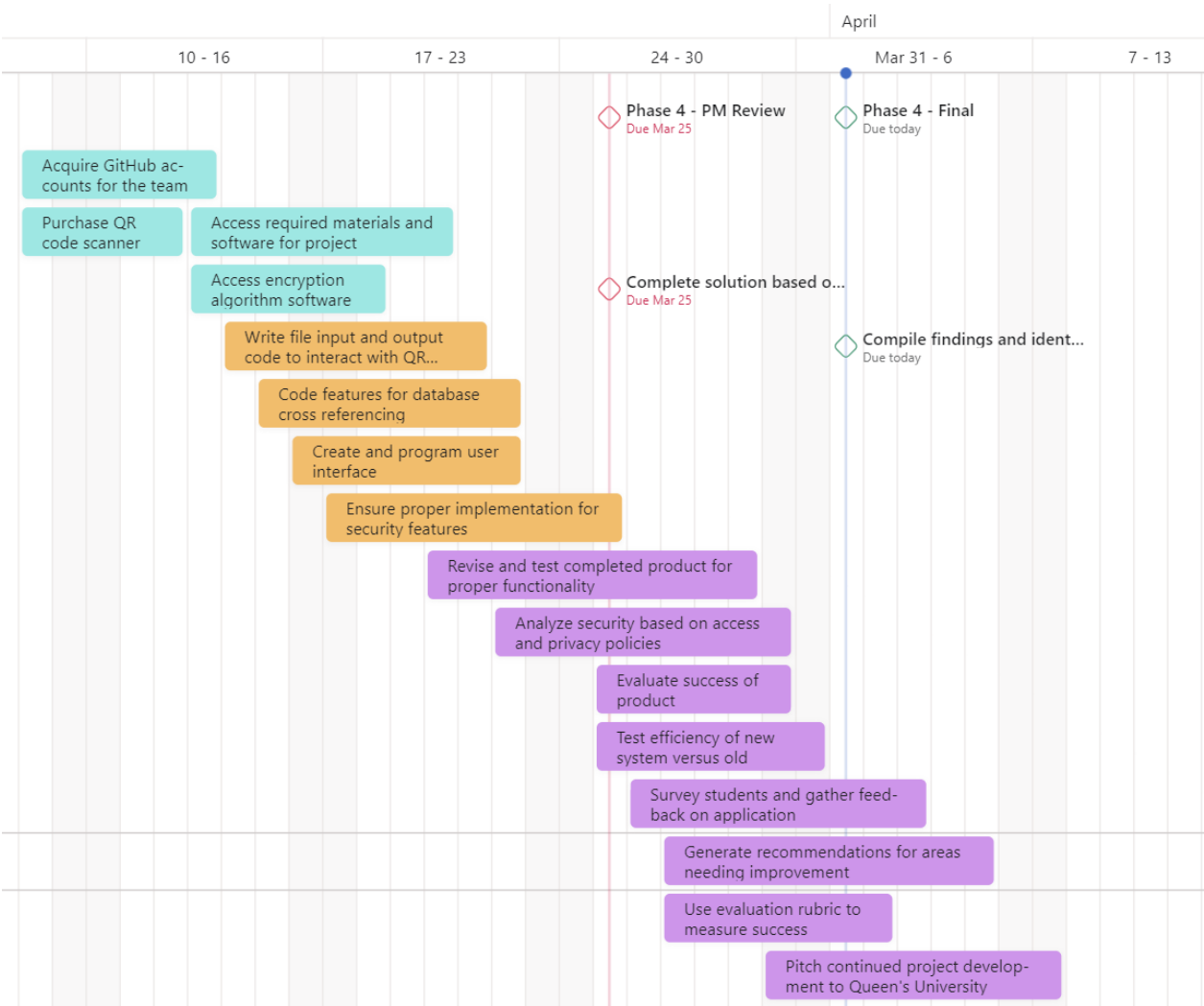


Figure 12: Updated project development Gantt Chart.

Table 4: List of locations for QR scanners.

Name of Location	Description	Amount of QR code scanners required
"Leonard Dinning Hall"	One of the main Queen's University Dining Areas	2
Ban Righ Dinning Hall	One of the main Queen's University Dining Areas	2
"Location 21" (David C. Smith)	Additional Dining Area	1
"Jean Royce Dining Hall"	One of the main Queen's University Dining Areas	1
"The Lazy Scholar" (Victoria Hall)	Additional Dining Area	2
"Starbucks" (Mitchel Hall)	Franchises	1
"Flipit" (Mackintosh-Corry Hall)	Additional Dining Area	1
"ARC Gym" (Queen's Center)	Queen's University Athletic and Recreation Center	2
"Goodes Café" (Goodes Hall)	Additional Dining Area	2
"The Library Café" (Stauffer Library)	Café	1
"Wally's" (Botterell Hall)	Additional Dining Area	1
"Tim Hortons" (Biosciences Building)	Franchises	1
"KHAO Pho Now" (Queen's Center)	Additional Dining Area	1
"Pita Pit" (Mackintosh-Corry Hall)	Franchises	2
"Peach" (New Medical Building)	Additional Dining Area	1
"Zapies" (Queen's Center)	Additional Dining Area	1
"Pizza Pizza" (Mackintosh-Corry Hall)	Additional Dining Area	2
"Buster Juice" (Queen's Center)	Franchises	1
"MC2" (Mackintosh-Corry Hall)	Additional Dining Area	1

Table 5: Project success evaluation rubric.

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Core Integration Functionality	All core functionalities work together flawlessly. Offers features that consider large scope - ex. dead phone and only has physical student card.	Integration/ final project of different systems may have minor issues or room for improvement.	Integration/ final project has significant issues or does not fully meet requirements.	Integration/final project is incomplete or not functional.
User Interface Design	A visually appealing and intuitive design that enhances the user experience.	A quality design that is user-friendly but may have minor usability issues.	Design is basic or lacks some user-friendly elements.	Poor design with significant usability issues.
Data Encryption	All sensitive data is properly encrypted.	Most sensitive data is encrypted, but there may be some gaps in encryption practices.	Little/ not working data encryption.	No data encryption.
Efficient Data Management	Fosters sustainability through seamless integration of databases. Working data management system, that allows access to and editing student information when needed.	Minor issues with data access and transfer.	Some working data storage system.	Not working/ functional data management.
QR Code Generation (Not evaluated)	QR Code generation algorithm to store student card information works flawlessly.	Generation of QR Code works well but has minor issues.	QR Code generation attempted but doesn't meet project goal of storing data within.	No QR Code generation functionality.

Table 6: Work distribution chart for document as well as project.

Task	Description of Activity	Activity Duration (hours)	Individual Responsible for Activity
Document structuring and formatting	Creating, proofreading, and organizing the document. Checking for proper use of acronyms, references, and other specified formatting.	2	Fedor
Executive Summary	Detailing information described in report with high level overview.	1	Rafael
1.1 Innovation Objectives	Stating the objectives of the innovation and introducing the business positioning statement.	1.5	Rafael
1.2 Background Information	Revising, condensing, and updating the information summary to accurately reflect the final product.	4	Fedor
1.3.1 Transaction Processing	Describing the general function of the QR scanner.	1	Fedor
1.3.2 Application Front-End	Describing the general functions and features of the program front-end as well as creating the user flowchart.	2	Fedor
1.3.3 Application Back-End	Describing the general function of the program back-end.	1	Fedor
1.3.4 Security and Encryption	Describing the general function of the security part and how it involves the end-user.	1	Daniil
1.3.5 Environmental and Societal Costs and Benefits	Detailing the societal and environmental costs and benefits associated with the innovation.	1.5	Rafael
1.3.6 Decommissioning Plans	Updating the decommissioning plans from previous report.	0.5	Fedor
1.4 Conclusions	Summarizing and synthesizing the information in Section 1.	1	Rafael
2.1 Potential Innovations and Decision Making	Describing potential innovations and selection process used.	2	Ray
2.2.1 Transaction Processing	Nikhil: Described the operation of the QR scanner. Fedor: Editing and revising draft.	Nikhil: 1 Fedor: 0.5	Nikhil & Fedor
2.2.2 Application Front-End	Nikhil: Described how the front-end of the application worked. Fedor: Editing and revising draft.	Nikhil: 2 Fedor: 2	Nikhil & Fedor
2.2.3 Application Back-End	Described how the back end of the application worked	3	Nikhil
2.2.4 Security and Encryption	Described how the security works, encryption and decryption methods.	3	Daniil
2.2.5 Environmental and Societal Costs and Benefits	Detailing the societal and environmental costs and benefits factors that were considered during development.	1	Rafael

Task	Description of Activity	Activity Duration (hours)	Individual Responsible for Activity
2.2.6 Modelling and Analysis	Describing the modelling used to inform the final prototype.	1	Fedor
2.3 Project Plan	Outlining updated project plan.	3	Ray
2.4 Financial Analysis	Described the financial part and calculated the costs.	1.5	Daniil
2.5 Evaluation	Evaluating the success of the innovation based on evaluation matrix and other metrics.	2	Rafael
QR scanning	Figured out the basics of the QR code scanner and making it successfully be able to interact with the Java GUI app	1.5	Nikhil
Program front-end layout	Made the basic layout and structure of the front-end including setting up the grid layout, adding headers and adding the background	5	Nikhil
Current date and time panels	Linked to libraries in JAVA and created a continuous timer that gets new info from the library every second.	3	Nikhil
Hourly and daily users	Created a linked list that is stored when the program is shut as well as creating a function that updates the number every time a code is scanned	3	Nikhil
Display of QR holder information	Display given info for each unique QR code as well as displaying error messages when a QR code was not scanned	1.5	Nikhil
Program database	Created structure and layout of the database including uploading photos and creating a csv layout	.5	Nikhil
Writing and reading from program database	Figured out how to read and write from the database to update info as codes were scanned	1	Nikhil
Program security	Created a security part of the project. Implemented encryption and decryption processes for data protection.	7	Daniil
Buttons and Flex field	Fedor: Programming the action listeners for mode buttons as well as a text field for the input of flex. Nikhil: Created a class for the rounded blue buttons.	Fedor: 1 Nikhil: 1	Fedor & Nikhil
Decision structures	Fully creating the valid flex amount input verification as well as the structure for determining which mode the program is in (but not actually coding what it does based off that information).	1	Fedor
Revisions	Revising Phase 4 report based off project manager feedback.	5	All